Light

GCSE Computer Science - Cyber Security Prevention (Set 2)

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GCSE • AQA 8525 • Grades 3—9 | Topic test | 30 min | 55 marks | Answered: 0 / 30

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← Back to Cyber Security Prevention

Multiple-choice topic test for **GCSE Computer Science (AQA 8525)**. You may skip questions. When you submit, you'll see your score, a grade estimate, and worked explanations. **Tip:** use the stopwatch above to time yourself.

Q1 (1 mark)

How does voice recognition authenticate a user?

- ○ **A.** By analysing unique characteristics of speech patterns, pitch, and vocal tract shape
- ○ **B.** By checking the volume of their voice
- ○ **C.** By counting how many words they say
- ○ **D.** By measuring how loudly they can shout

Q2 (1 mark)

Why might 'correct horse battery staple' be a stronger password than 'Tr0ub4dor&3'?

- ○ **A.** Because it contains numbers
- ○ **B.** Because it uses special characters
- ○ **C.** Because its greater length provides more possible combinations, making it harder to crack despite using only lowercase letters
- ○ **D.** Because it is easier to type quickly

Q3 (1 mark)

Why do some websites offer audio CAPTCHA as an alternative to image-based CAPTCHA?

- ○ **A.** Because audio is louder
- ○ **B.** To make the website more musical
- ○ **C.** Because audio loads faster than images
- ○ **D.** To allow users with visual impairments to complete the verification process

Q4 (1 mark)

Why do email confirmation links typically expire after 24-48 hours?

- ○ **A.** To limit the window during which an intercepted link could be misused by an attacker
- ○ **B.** To save storage space on email servers
- ○ **C.** Because email technology cannot support longer links

- ○ **D.** To encourage users to check email more frequently

Q5 (1 mark)

What is a 'zero-day' vulnerability?

- ○ **A.** A bug that only appears on the first day of each month
- ○ **B.** A feature that was added on day zero of development
- ○ **C.** A vulnerability that takes zero days to fix
- ○ **D.** A security flaw that is being exploited before the software vendor knows about it or has released a fix

Q6 (1 mark)

What are backup codes used for in two-factor authentication systems?

- ○ **A.** To make the account more colourful
- ○ **B.** To back up your files automatically
- ○ **C.** To allow account access if you lose your primary second factor device, such as your phone
- ○ **D.** To create copies of your password

Q7 (1 mark)

What does 'quarantine' mean in anti-malware software?

- ○ **A.** Permanently deleting the infected file immediately
- ○ **B.** Sending the file to the software company
- ○ **C.** Isolating a suspicious file so it cannot run or spread while allowing later review or recovery
- ○ **D.** Making the file run faster

Q8 (1 mark)

Why might a firewall block outgoing (outbound) network connections as well as incoming ones?

- ○ **A.** To prevent malware on the computer from sending stolen data to attackers or communicating with command servers
- ○ **B.** To make the internet connection faster
- ○ **C.** To stop electricity flowing out of the computer
- ○ **D.** To reduce the weight of data leaving the network

Q9 (1 mark)

What is the purpose of account lockout after multiple failed login attempts?

- ○ **A.** To punish users who forget their passwords
- ○ **B.** To encourage users to create new accounts
- ○ **C.** To save electricity by reducing login attempts

○ **D.** To prevent brute-force attacks where attackers systematically try many password combinations

Q10 (1 mark)

What is stored when you register your fingerprint with a biometric system?

○ **A.** A photograph of your entire hand

○ **B.** A mathematical template derived from fingerprint features, not an actual image of the fingerprint

○ **C.** Your fingerprint in a glass case

○ **D.** A video recording of you placing your finger on the scanner

Q11 (1 mark)

How does heuristic detection in anti-malware software differ from signature-based detection?

○ **A.** Heuristic detection only works on Tuesdays

○ **B.** There is no difference - they are the same thing

○ **C.** Heuristic detection is slower because it uses paper records

○ **D.** Signature detection matches known malware patterns, while heuristic detection analyses behaviour to identify suspicious activities that might indicate new, unknown threats

Q12 (1 mark)

Why should employees report suspected social engineering attempts even if they did not fall for them?

- ○ **A.** Because reporting helps the organisation identify ongoing attacks, warn other potential targets, and improve training and defences
- ○ **B.** To get colleagues into trouble
- ○ **C.** To fill up the IT department's inbox
- ○ **D.** Because reporting earns holiday time

Q13 (2 marks)

Some websites use 'invisible CAPTCHA' that analyses user behaviour without displaying puzzles. How does this approach verify users are human?

- ○ **A.** By reading users' minds
- ○ **B.** By measuring the temperature of the user's hands
- ○ **C.** By checking if users are wearing glasses
- ○ **D.** By analysing mouse movements, typing patterns, browsing behaviour, and interaction timing - humans show natural variations that bots typically cannot replicate convincingly

Q14 (2 marks)

What is biometric spoofing, and why does it matter for security?

- ○ **A.** Spoofing is when attackers create fake biometric samples (like fingerprint moulds or printed photos) to fool scanners - it matters because successful spoofing bypasses authentication meant to prove physical presence of the authorised person

- ○ **B.** Spoofing means the biometric scanner is broken
- ○ **C.** Spoofing is when users forget their biometrics
- ○ **D.** Spoofing makes biometric scanners work faster

Q15 (2 marks)

Why are password hints considered a security weakness, and why have many services removed them?

- ○ **A.** Because hints make passwords too long
- ○ **B.** Because hints often provide enough information for attackers to guess passwords - especially combined with social media research - effectively creating a backdoor that bypasses password strength
- ○ **C.** Because hints are too difficult to create
- ○ **D.** Because hints slow down the login process

Q16 (2 marks)

What is a staged rollout for software updates, and why do large organisations use this approach?

- ○ **A.** Updates are deployed first to a small group of test users, then gradually expanded to larger groups - this catches problems before they affect everyone while still applying updates promptly
- ○ **B.** Staged rollout means never installing updates
- ○ **C.** Staged rollout means installing all updates at once
- ○ **D.** It refers to theatrical performances about software updates

Q17 (2 marks)

Some services send clickable confirmation links while others send numeric verification codes to type in. What security consideration might influence this choice?

- ○ **A.** Links are always more secure than codes
- ○ **B.** Codes are always more secure than links
- ○ **C.** Verification codes require the user to actively return to the legitimate site to enter them, avoiding the risk of clicking a phishing link embedded in a fake confirmation email
- ○ **D.** The choice is purely random with no security implications

Q18 (2 marks)

How do hardware security keys like YubiKey provide stronger authentication than SMS codes?

- ○ **A.** They use cryptographic challenge-response that cannot be intercepted or phished - the key must be physically present and only responds to legitimate site requests
- ○ **B.** They are larger and heavier
- ○ **C.** They are made of more expensive materials
- ○ **D.** They glow in the dark

Q19 (2 marks)

What problems can arise from anti-malware false positives, and how should they be handled?

- ○ **A.** False positives have no negative effects

- **B.** False positives make computers run faster
- **C.** Legitimate files incorrectly flagged as malware may be quarantined or deleted, disrupting operations - users should verify detections before permanent deletion and report false positives to software vendors
- **D.** False positives only affect files that do not matter

Q20 (2 marks)

A caller claims to be conducting an IT security audit and asks an employee for their login credentials to 'verify the account is properly secured'. What should the employee do?

- **A.** Provide the credentials immediately to help with the audit
- **B.** Provide only the password but not the username
- **C.** Ask the caller to wait while getting a colleague's credentials instead
- **D.** Refuse to provide credentials, as legitimate security audits never require users to reveal passwords - verify any audit through official internal channels before participating

Q21 (2 marks)

What environmental factors can affect the reliability of fingerprint scanners?

- **A.** The colour of the room walls
- **B.** The number of people nearby
- **C.** Wet, dirty, or damaged fingers can produce poor scans; extreme cold can affect finger circulation; oils, lotions, and abrasions

can alter fingerprint appearance - affecting recognition accuracy

○ **D.** The day of the week

Q22 (2 marks)

How do application-layer firewalls differ from traditional packet-filtering firewalls?

○ **A.** Application firewalls only work with mobile apps

○ **B.** There is no difference between them

○ **C.** Application firewalls are physically larger devices

○ **D.** Packet-filtering firewalls examine basic header information like IP addresses and ports, while application-layer firewalls understand specific protocols and can inspect actual content, detecting threats hidden within allowed traffic

Q23 (2 marks)

What is update rollback, and why is this capability important for system stability?

○ **A.** Rollback means updates happen automatically

○ **B.** Rollback allows reverting to the previous software version if an update causes problems - providing a recovery path when patches introduce bugs, compatibility issues, or system instability

○ **C.** Rollback prevents any updates from being installed

○ **D.** Rollback makes computers run backwards

Q24 (3 marks)

A fitness centre implements these security measures for its online membership system: (1) CAPTCHA on the account creation page; (2) Email verification before account activation; (3) Account lockout after five failed login attempts. Analyse how each measure protects the system and its members.

- ○ **A.** CAPTCHA prevents automated mass account creation that could be used for spam or fraud; email verification ensures accounts are linked to real, accessible email addresses preventing impersonation; account lockout stops password guessing attacks protecting member accounts from unauthorised access - together they secure the registration and login processes
- ○ **B.** These measures only slow down legitimate members
- ○ **C.** Only the CAPTCHA provides any real protection
- ○ **D.** Fitness centres do not need security measures

Q25 (3 marks)

A restaurant chain discovers that point-of-sale terminals at multiple locations were compromised, with customer payment card data stolen. Investigation reveals: (1) All terminals used the same default administrator password; (2) Antivirus software was installed but not updated for two years; (3) Staff had no training on recognising phishing emails. Analyse how these failures enabled the breach.

- ○ **A.** The payment system was inherently insecure and nothing could have prevented the breach
- ○ **B.** Restaurant systems cannot be targeted by cyber attacks
- ○ **C.** Only the default password mattered - the other issues were irrelevant
- ○ **D.** Shared default passwords meant that compromising one terminal's credentials gave access to all locations; outdated antivirus failed to detect malware that had signatures added after the last

update; untrained staff likely opened phishing emails that delivered the initial malware - each weakness contributed to both the initial compromise and its spread

Q26 (3 marks)

An online gaming platform offers players several account protection options: (1) Strong password requirements; (2) Optional two-factor authentication with authenticator app; (3) Email notifications for login from new devices; (4) Account recovery requiring identity verification. Analyse how these measures protect player accounts and in-game assets.

- ○ **A.** Gaming accounts do not need protection since games are not serious
- ○ **B.** Strong passwords resist guessing attacks; 2FA prevents access even if passwords are compromised; login notifications alert owners to unauthorised access attempts; verified recovery prevents attackers from hijacking accounts through fake recovery requests - together protecting both accounts and valuable in-game purchases, progress, and virtual items
- ○ **C.** Only the password requirements provide protection
- ○ **D.** These measures make accounts less secure

Q27 (3 marks)

A public library provides computers for visitor use. They implement: (1) Session timeouts after 30 minutes of inactivity; (2) Automatic clearing of browser data when users log out; (3) Deep freeze software that restores computers to a clean state on restart; (4) Prominent signs reminding users to log out of personal accounts. Analyse the security purposes these measures serve.

- ○ **A.** These measures only inconvenience library users

○ **B.** Session timeouts prevent abandoned sessions being accessed by later users; browser clearing removes passwords, history and cached data between users; deep freeze eliminates any malware or changes from each session; logout reminders reduce users accidentally leaving accounts accessible - together protecting both the library systems and users' personal accounts

○ **C.** Libraries do not need security on public computers

○ **D.** Only the deep freeze software provides any protection

Q28 (3 marks)

An accounting firm protects client financial data with: (1) Unique user accounts for each employee with role-based permissions; (2) Multi-factor authentication required for remote access; (3) Automatic screen locking after 5 minutes of inactivity; (4) Audit logs recording all access to client files. Analyse the security and compliance purposes these measures serve.

○ **A.** Individual accounts with role-based access ensure employees only reach data necessary for their role, limiting both accidental and malicious exposure; MFA protects remote access where network security cannot help; screen locking prevents opportunistic access when employees step away; audit logs create accountability and support regulatory compliance by tracking all data access

○ **B.** These measures reduce the firm's productivity without security benefit

○ **C.** Only multi-factor authentication provides meaningful protection

○ **D.** Accounting firms do not handle sensitive data

Q29 (3 marks)

A school proposes using fingerprint scanners for library book checkout and lunch payment. Some parents raise concerns. Analyse both the

potential benefits and the privacy considerations specific to collecting biometric data from children.

- ○ **A.** There are no privacy considerations with children's biometric data
- ○ **B.** Benefits include faster checkout, reduced lost cards, and preventing lunch money theft; however, children cannot meaningfully consent to permanent biometric collection, the data cannot be changed if compromised, long-term storage creates breach risks, and using biometrics for low-security purposes may normalise surveillance - requiring careful consideration of whether benefits justify collecting sensitive data from minors
- ○ **C.** Fingerprint collection from children is identical to collecting it from adults
- ○ **D.** Schools should never use any technology

Q30 (3 marks)

A company discovers that despite having automatic updates enabled, some computers have not received security patches for months. Investigation reveals different causes for different machines. Analyse what factors might cause automatic updates to fail silently.

- ○ **A.** Automatic updates never fail once enabled
- ○ **B.** Update failures only affect old computers
- ○ **C.** If updates fail, the computer will always display a clear warning
- ○ **D.** Updates may fail due to: insufficient disk space preventing download or installation; computers being powered off during scheduled update times; network restrictions blocking update servers; corrupted update components; third-party software conflicts; group policy misconfiguration - demonstrating that automatic updates require monitoring to ensure they actually succeed

Total: **55** marks • Time: **30 min**

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